

Comparing LLM Text Annotation Skills: A Study on Human Rights Violations in Social Media Data

Step 1: Dataset Gathering and Sampling

- Gather social media posts from publicly available Telegram channels related to the war between Russia and Ukraine, in Russian (966 posts) and to certain extent in Ukrainian (34 posts).
- Extract 1000 observations out of a large dataset consisting of Human Rights Violations (HRV) to form a benchmark.
- Make sure the sample is moderately imbalanced in class: 517 positive (presence of HRV) and 483 negative (absence of unclear HRV).

Step 2: Annotating and Building Gold Standard

- Designate two proficient native speakers, fluent in both Russian and Ukrainian, to independently evaluate each post for a binary classification regarding the presence or absence of references to HRV.
- Calculate inter-annotator agreement through Cohen's Kappa, for those 184 posts related to disagreement, bring in another expert annotator to decide and resolve labels and construct the final gold-standard dataset.

Step 3: Model Selection

- Choose five LLMs for evaluation: GPT-3.5 Turbo and GPT-4.0 (proprietary), Claude-2 (proprietary with safety focus), LLaMA-3.2-1B (open-source), and Mistral-7B (open-source).
- Cherry-pick models to strike a balance between high-performance proprietary models and resource-efficient open-source models, without any fine-tuning.

Step 4: Experimental Arrangement

- All experiments have to be executed in the Python environment within Google Colab on high-RAM A100 hardware to promote accessibility and reproducibility.
- Set stable random seeds for all iterations to ensure identical results.
- Do not fine-tune; only ground models' zero-shot and few-shot capacities to model realistic scenarios under few-data sets.

Step 5: Instructional Design and Implementation Create prompts in English and Russian to test linguistic correspondence to the dataset.

Incorporate two prompting procedures:

- a) Zero-shot: Provide explicit advice without exemplary cases, testing pre-trained generalization.
- b) Few-shot: Accompany instructions with some marked instances to capitalize on in-context learning.

apply these prompts to each model, iterating through the 1000-post dataset to generate annotations (binary labels for presence of HRV).

Step 6: Performance Appraisal

- Compare generated LLM labels to the human gold standard.
- Estimate statistics like precision, recall, F1 score, and accuracy, and emphasize class imbalance through balance.
- Analyzing results independently for sets: 816 posts agreed upon by fullhumans and 184 having initial disagreement (ablation study to check if models face an identical set of challenges as humans).

Step 7: Error Analysis and Checks for Reproducibility

- Discuss error types (such as false positives from implied events, false negatives from indirect reference) and differences in models.
- Put all prompts, seeds, and hyperparameters in an openly viewable GitHub repository to allow for full reproducibility.